

Photon Theory

- Intensity (I) = $\frac{\text{Power}}{\text{Area}} = \frac{E}{tA}$
= energy per unit area per unit time
- Point source $I = \frac{P}{4\pi r^2} \Rightarrow I \propto \frac{1}{r^2}$
- Line source $I = \frac{P}{2\pi r l} \Rightarrow I \propto \frac{1}{r}$

- no. of Photons
 $E = n h\nu \Rightarrow n = \frac{E}{h\nu} = \frac{E\lambda}{hc}$
- no. of Photons per unit time
 $\frac{n}{t} = \frac{E}{th\nu} = \frac{P}{h\nu} = \frac{IA}{hc}$
- no. of Photons per area per unit time
 $\frac{n}{At} = \frac{E}{tAh\nu} = \frac{P}{Ah\nu} = \frac{I\lambda}{hc}$

- Power of incident radiation (P)

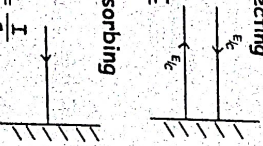
$$P = \frac{n h c}{\lambda} \quad P \propto \left(\frac{n}{\lambda}\right)$$

source 1 $P_1 \rightarrow \lambda_1 \rightarrow n_1$
source 2 $P_2 \rightarrow \lambda_2 \rightarrow n_2$

$$\frac{P_1}{P_2} = \frac{n_1}{n_2} \times \frac{\lambda_2}{\lambda_1}$$

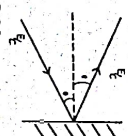
Dual Nature of Radiation

- Momentum of photon (p) = $\frac{E}{c} = \frac{h}{\lambda}$
- Force (F) = $\frac{\Delta p}{\Delta t}$
- Radiation Pressure = $-\frac{F}{A}$
- For perfectly reflecting surface
 $\Delta p = \frac{2E}{c} \quad F = \frac{2P}{c}$
- Rad. Pressure = $\frac{2I}{c}$
- For Perfectly Absorbing Surface
 $\Delta p = \frac{E}{c} \quad F = \frac{P}{c}$
- Rad. Pressure (P_r) = $\frac{I}{c}$



DUAL NATURE OF RADIATION & MATTER

- Perfectly reflecting at an angle
 $\Delta p = \frac{2E}{c} \cos \theta$
 $F = \frac{2P}{c} \cos \theta$
Rad. Pressure = $\frac{2I}{c} \cos \theta$

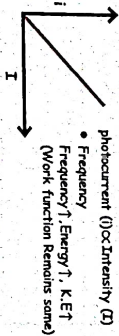


- PHOTOELECTRIC EFFECT
- Energy of photon $E = h\nu$
- ν = Frequency of incident light in Hz = $\frac{c}{\lambda}$
- Max. kinetic energy of emitted photoelectron
 $(K.E.)_{\text{max}} = E - w = \frac{1}{2} m v_{\text{max}}^2$

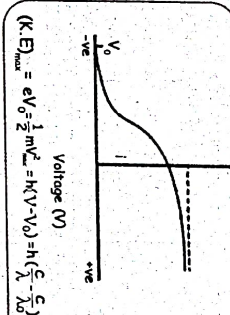
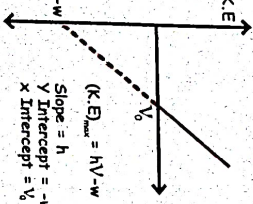
Work function (w)

- Minimum energy required for photoelectric effect to occur
 $w = h\nu_0 = h \frac{c}{\lambda_0} \quad h = 6.63 \times 10^{-34} \text{ Js}$
- ν_0 = Threshold frequency in Hz
 λ_0 = Threshold wavelength in m
- Work function only depends on nature of metal

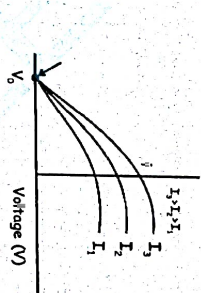
- Factors affecting photoelectric effect
- Intensity $\uparrow$, photoelectrons $\uparrow$, photocurrent $\uparrow$ (K.E. Remains same)



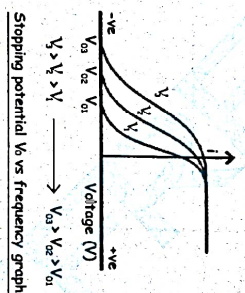
- Anode potential Opposes K.E. of electron
- Max Negative anode potential = Stopping potential (V_0) for which Photocurrent (I) = 0



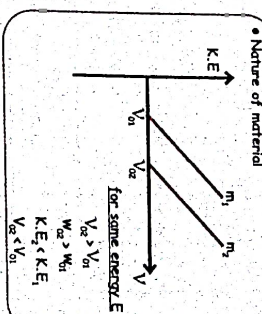
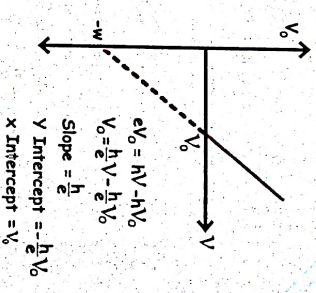
- Factors affecting stopping potential
- Intensity $\uparrow$, K.E. Remains same
- Stopping potential remains same



- Frequency $\uparrow$, Energy $\uparrow$, K.E. $\uparrow$, $V_0 \uparrow$



- Stopping potential V_0 vs frequency graph
 $V_0 > V_0' \rightarrow V_0 > V_0' > V_0$



- Nature of material
- Conceptual question
If green color have just sufficient energy for photoelectric effect
Increasing order of λ : V, X, B, G, Y, O, R
no photo electric effect
same photo electric effect
 $E \uparrow \downarrow \lambda \uparrow$

- Useful conversions
- Wave length (nm) $\rightarrow$ K.E (eV)
 $K.E = 1240 \text{ (eV)}$
- Wave length (λ) $\rightarrow$ K.E (eV)
 $K.E = 1240 \text{ (eV)}$

- Two identical photo cathode receive light of frequencies f_1 & f_2 . If velocity of photo electrons are v_1 & v_2 then $v_1^2 - v_2^2 = \frac{2h}{m} (f_1 - f_2)$

Dual Nature of Matter

- Wave nature of Matter
- De Broglie waves
- fast moving particles like electron with much smaller mass behaves like a wave i.e., Circular stationary waves
- $E = mc^2 = \frac{h\nu}{\lambda}$
 $\lambda = \frac{h}{mv} = \frac{h}{m\gamma v} = \frac{h}{\gamma m_0 v}$
 $K.E = qV$ (for charged particle)
 $K.E = \frac{1}{2} K_T$ (thermal neutron)
 K_B = Boltzmann's constant
 T = Temperature in Kelvin

- Thermal Neutron
 $\lambda = \frac{h}{\sqrt{3mk_B T}} = \frac{30.83 \text{ \AA}}{\sqrt{T}}$

- Electron $\lambda = \frac{12.27 \text{ \AA}}{\sqrt{V}}$

- Proton $\lambda = \frac{0.286 \text{ \AA}}{\sqrt{V}}$

- Deuteron (λ) = $\frac{0.202 \text{ \AA}}{\sqrt{V}}$

- α - Particle (λ) = $\frac{0.101 \text{ \AA}}{\sqrt{V}}$

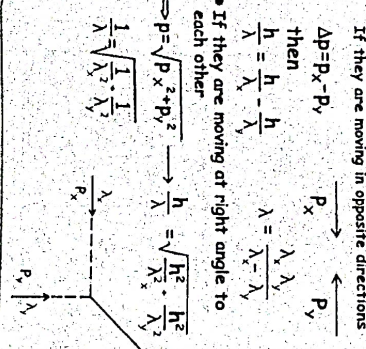
- ^1H (Proton) $\rightarrow$ 1 Proton $\rightarrow$ m, q
- ^2H (Deuteron) $\rightarrow$ 1 Proton + 1 Neutron $\rightarrow$ 2m, q
- ^4He (α Particle) $\rightarrow$ 2 Proton + 2 Neutron $\rightarrow$ 4m, 2q

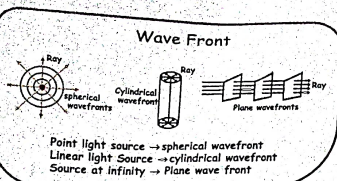
- Relationship b/w wavelength of photon & that of electron
- Ratio of wavelength of photon to that of electron with same energy E
 $\frac{\lambda_{\text{photon}}}{\lambda_e} = \frac{c \sqrt{2m}}{E} \quad \lambda_{\text{photon}} \propto \lambda_e^2$

- Ratio of K.E. of electron to that of photon with same wavelength for Same
 $\frac{K.E.}{K.E._{\text{photon}}} = \frac{V}{2c}$

- A particles formed due to completely inelastic collision of particle X' and Y' having de Broglie wave length λ_X and λ_Y respectively.

- If they are moving in opposite directions
 $\Delta p = p_X - p_Y$
then
 $\lambda = \frac{h}{\Delta p} = \frac{h}{p_X - p_Y} = \frac{\lambda_X \lambda_Y}{\lambda_X - \lambda_Y}$
- If they are moving at right angle to each other
 $\Rightarrow p = \sqrt{p_X^2 + p_Y^2} \rightarrow \frac{h}{\lambda} = \sqrt{\frac{h^2}{\lambda_X^2} + \frac{h^2}{\lambda_Y^2}}$
 $\frac{1}{\lambda} = \sqrt{\frac{1}{\lambda_X^2} + \frac{1}{\lambda_Y^2}}$

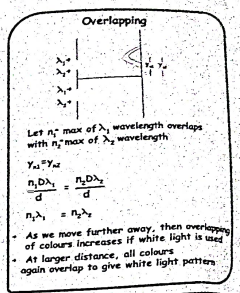




Huygen's principle

i) Each point on a wavefront acts as a fresh source of new disturbance, called secondary waves or wavelets.
 The secondary wavelets spread out in all directions with the speed of light in the given medium.
 ii) A common envelope or common tangent to these secondary wavelets at any later time gives secondary wavefront at that time.

WAVE OPTICS



	Incident wavefront	Reflected wavefront
Concave Mirror	Plane wavefront	Spherical converging wavefront
Convex Mirror	Plane wavefront	Spherical diverging wavefront
	Incident wavefront	Refracted wavefront
Convex Lens	Plane wavefront	Spherical converging wavefront
Concave Lens	Plane wavefront	Spherical diverging wavefront

Phase Difference & Path Difference

$$\Phi = \frac{2\pi}{\lambda} \Delta x$$

Phase Difference & Time Difference

$$\Phi = \frac{2\pi}{T} \Delta t$$

Resultant Amplitude

$$Y_1 = A_1 \sin \omega t \text{ and } Y_2 = A_2 \sin (\omega t + \Phi)$$

Resultant $A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \Phi}$

- $\cos \Phi = 1 \Rightarrow A = A_{\text{max}} = \sqrt{(A_1 + A_2)^2} = A_1 + A_2$
- $\cos \Phi = -1 \Rightarrow A = A_{\text{min}} = \sqrt{(A_1 - A_2)^2} = A_1 - A_2$

$\frac{A_{\text{min}}}{A_{\text{max}}} = \frac{A_1 - A_2}{A_1 + A_2}$
 • Intensity $\propto (\text{amplitude})^2$

YDSE in Liquid

When YDSE setup is immersed in a liquid, there is change in wavelength

$$n = \frac{c}{v} = \frac{v\lambda}{v\lambda'} = \frac{\lambda}{\lambda'} \quad n \rightarrow \text{refractive index}$$

$$\lambda_{\text{medium}} = \lambda' = \frac{\lambda}{n} \quad \text{or} \quad \lambda' = \frac{\lambda}{\mu} \quad \mu \rightarrow \text{refractive index}$$

In air $v_1 = n \frac{d\lambda}{d}$
 In medium $v_2 = n \frac{d\lambda'}{d} = \frac{nd\lambda}{d\mu}$

Fringe width in air $\beta = \frac{d\lambda}{d}$
 In medium $\beta' = \frac{d\lambda'}{d\mu} \Rightarrow \beta_{\text{med}} = \frac{\beta}{\mu}$

Resultant Intensity

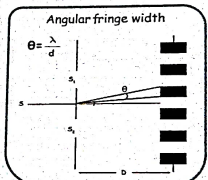
We have, $I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \Phi$

- $\cos \Phi = 1 \Rightarrow I = I_{\text{max}}$
- $I_{\text{max}} = I_1 + I_2 + 2\sqrt{I_1 I_2}$
- $I_{\text{max}} = (\sqrt{I_1} + \sqrt{I_2})^2$
- $\cos \Phi = -1 \Rightarrow I = I_{\text{min}}$
- $I_{\text{min}} = (\sqrt{I_1} - \sqrt{I_2})^2$
- $\frac{I_{\text{min}}}{I_{\text{max}}} = \frac{(\sqrt{I_1} - \sqrt{I_2})^2}{(\sqrt{I_1} + \sqrt{I_2})^2}$
- $I_{\text{min}} \propto A_{\text{min}}^2$ & $I_{\text{max}} \propto A_{\text{max}}^2$
- $\frac{I_{\text{min}}}{I_{\text{max}}} = \frac{A_{\text{min}}^2}{A_{\text{max}}^2} = \left(\frac{A_1 - A_2}{A_1 + A_2}\right)^2$
- If $I_1 = I_2 = I_0$
 $\Rightarrow I = 4I_0 \cos^2 \frac{\Phi}{2}$

Young's Double-slit experiment (YDSE)

Path difference $\Delta x = \frac{Yd}{D}$
 In general $\Delta x = \frac{Yd}{D}$
 Distance of Minima and Maxima from Central maximum

Maxima	Minima
$Y_n = \frac{nD\lambda}{d} \quad n = 0, 1, 2, \dots$	$Y_n = (2n-1) \frac{D\lambda}{2d} \quad n = 1, 2, \dots$



Constructive interference	Destructive interference
Phase difference at the point of observation $\Phi = 0^\circ$ or $2n\pi, n = 0, 1, 2, \dots$ Also, $\Delta x = n\lambda, n = 0, 1, 2, \dots$ Resultant intensity at the point of observation is maximum $I = I_{\text{max}} = (\sqrt{I_1} + \sqrt{I_2})^2$	$\Phi = 180^\circ$ or $\Phi = (2n-1)\pi; n = 1, 2, \dots$ Also, $\Delta x = (2n-1) \frac{\lambda}{2}, n = 1, 2, 3, \dots$ Resultant intensity at the point of observation will be minimum $I = I_{\text{min}} = (\sqrt{I_1} - \sqrt{I_2})^2$

Intensity at any point on screen

For all maxima $I = 4I_0$ (If $I_1 = I_2 = I_0$)
 For all minima, $I = 0$
 Note:
 Fringe visibility $V = \frac{I_{\text{max}} - I_{\text{min}}}{I_{\text{max}} + I_{\text{min}}}$

Introduction Of Thin Transparent Sheet in YDSE

Optical path length and geometrical path length

Refractive index $\mu = \frac{c}{v}$

$$\mu = \frac{v\lambda}{v\lambda'} \Rightarrow \lambda_{\text{med}} = \frac{\lambda}{\mu}$$

Time taken by light to travel x length in medium.

$$t = \frac{x}{c/\mu} = \frac{\mu x}{c}$$

Distance travelled by light in vacuum in same time = optical path length

Optical Path Length (OPL) = velocity $\times$ time $= c \times \frac{\mu x}{c} = \mu x$

If Geometrical Path Length (GPL) = x , then
 $\text{OPL} = \mu x$, where μ is the refractive index of the medium

Fringe Width or Band width (β)

$$\beta_{\text{dark}} = \frac{D\lambda}{d}$$

$$\beta_{\text{bright}} = \frac{D\lambda}{d}$$

For interference pattern $\beta_{\text{dark}} = \beta_{\text{bright}} = \frac{D\lambda}{d}$

Introduction of thin transparent sheet

Path difference $\Delta x = s_2P - s_1P$

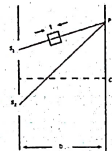
Additional path difference = $(\mu - 1)t$

Geometrical path difference before inserting sheet, $\Delta x = \frac{y^2}{2d}$

$$y = \frac{D}{d} \Delta x$$

After introducing sheet, $y' = \frac{D}{d} [\Delta x + (\mu - 1)t]$

Shift $S = y' - y = \frac{D}{d} (\mu - 1)t$



Interference of reflected light :
For normal incidence $n = 0$ so, $2\mu t = (2n - 1) \frac{\lambda}{2}$
Interference of refracted light :
For normal incidence, $2\mu t = n\lambda$

Single Slit Diffraction

Path difference = $\Delta x = d \sin \theta$

• Formation of first secondary minima

$$\text{Path difference} = \frac{\lambda}{2}$$

• Formation of 2nd secondary minima

Path difference = 2λ

• Formation of n^{th} secondary minima

$$d \sin \theta_n = n\lambda$$

$n = 1, 2, 3, \dots$

First secondary maxima

But the intensity of 1st secondary maxima is lower than central maximum

N^{th} secondary Maxima

$$x = (2n + 1) \frac{\lambda}{2}$$

$$n = 1, 2, 3, \dots$$

$$d \sin \theta_n = (2n + 1) \frac{\lambda}{2}$$

Ratio of intensities of central maxima and secondary maxima

$$1 : \frac{1}{2} : \frac{1}{61} : \frac{1}{121} : \dots$$

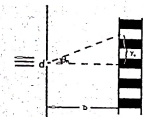
Distance of N^{th} secondary maxima from CM

$$\theta_n = (2n + 1) \frac{\lambda}{2d}$$

$$n = 1, 2, 3, \dots$$

$$y = (2n + 1) \frac{D\lambda}{2d}$$

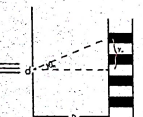
$$n = 1, 2, 3, \dots$$



Distance of N^{th} secondary minima from CM

$$y = \frac{nD\lambda}{d}$$

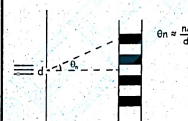
$$n = 1, 2, 3, \dots$$



WAVE OPTICS

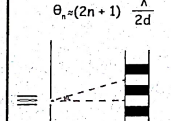
Angular position of N^{th} secondary minima

$$\theta_n \approx \frac{n\lambda}{d}$$

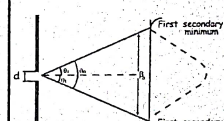


Angular position of N^{th} secondary Maxima

$$\theta_n = (2n + 1) \frac{\lambda}{2d}$$



Angular width and linear width of central maximum



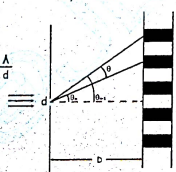
$$\beta_0 = \theta_0 D = \frac{2D\lambda}{d}$$

Angular width and linear width of secondary maxima

Angular position of n^{th} minimum, $\theta_n = n \frac{\lambda}{d}$

Angular position of $(n+1)^{\text{th}}$ minimum, $\theta_{n+1} = (n+1) \frac{\lambda}{d}$

$$\text{Linear width, } \beta = \frac{D\lambda}{d}$$



Angular width and Linear width of secondary minima

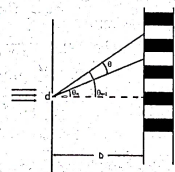
Angular position of n^{th} maximum $\theta_n = (2n + 1) \frac{\lambda}{2d}$

Angular position of $(n+1)^{\text{th}}$ maximum $\theta_{n+1} = (2(n+1) + 1) \frac{\lambda}{2d}$

$$\theta = \frac{\lambda}{d}$$

Linear width $\beta = \theta D$

$$\beta = \frac{D\lambda}{d}$$



Law of Malus

$$I \propto \cos^2 \theta$$

$$I = I_0 \cos^2 \theta$$

When $\theta = 0^\circ$ or 180° ,

$$\cos \theta = \pm 1 \Rightarrow I = I_0$$

When $\theta = 90^\circ$,

$$\Rightarrow \cos \theta = 0 \Rightarrow I = 0$$

Polarisation by Reflection

Brewster found that at the polarising angle, the reflected and transmitted rays are perpendicular to each other

$$n = \tan i_p$$

This is Brewster's Law

ATOMS

RUTHERFORD'S ATOM MODEL

- Majority of α -particles passed without any deviation.
- Some are scattered at small angle θ (impact parameter is equal to that of nuclear radius)
- Only few α particle retrace the path (impact parameter = 0)



BOHR ATOM MODEL

First postulate

$$F = \frac{1}{4\pi\epsilon_0} \frac{Ze^e}{r^2}$$

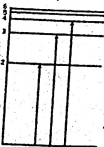
Second postulate

$$mvr = \frac{nh}{2\pi}$$

$$\frac{mv^2}{r} = \frac{1}{4\pi\epsilon_0} \frac{Ze^2}{r^2}$$

HYDROGEN SPECTRUM

Absorption spectrum



Electrons absorb only those photons whose energy = Energy difference of 2 shells

If atomic excitation takes place upto n^{th} shell starting from ground state then $(n-1)$ different photons are absorbed

EMISSION SPECTRUM

Total no. of different wavelength photons in emission spectrum = $\frac{n(n-1)}{2}$

All absorbed photons are emitted in emission spectrum

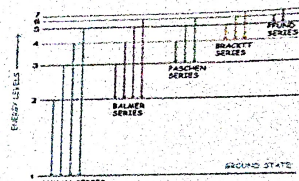
Wavelength of emitted photon

$$\frac{1}{\lambda} = R Z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

$$R = 10^3 \text{ cm}^{-1}$$

$$R = 10^7 \text{ m}^{-1}$$

LINE SPECTRUM OF HYDROGEN ATOM

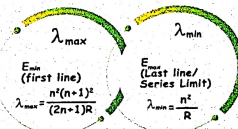


ENERGY

$$\text{Total energy} = -13.6 \frac{Z^2}{n^2} \text{ eV}$$

$$\text{K.E} = -\text{T.E} = +13.6 \frac{Z^2}{n^2} \text{ eV}$$

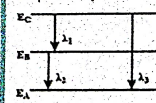
$$\text{P.E} = 2 \text{ T.E}$$



LINE SPECTRUM OF HYDROGEN ATOM

Spectral series	n_1	n_2	Wavelength	λ_{min} ($n_2 = n_1 + 1$)	λ_{max} ($n_2 = \infty$)	Region	Range
Lyman	1	2, 3, 4	$\frac{1}{\lambda_{12}} = R \left(\frac{1}{1^2} - \frac{1}{n^2} \right)$	$\frac{4}{3R}$	$\frac{1}{R}$	Ultra - violet	911.6 Å to 1216 Å
Balmer	2	3, 4, 5	$\frac{1}{\lambda_{23}} = R \left(\frac{1}{2^2} - \frac{1}{n^2} \right)$	$\frac{36}{5R}$	$\frac{4}{R}$	Visible	3646 Å to 6563 Å
Paschen	3	4, 5, 6	$\frac{1}{\lambda_{34}} = R \left(\frac{1}{3^2} - \frac{1}{n^2} \right)$	$\frac{144}{7R}$	$\frac{9}{R}$	Near infra-red	8204 Å to 18753 Å
Brackett	4	5, 6, 7	$\frac{1}{\lambda_{45}} = R \left(\frac{1}{4^2} - \frac{1}{n^2} \right)$	$\frac{400}{9R}$	$\frac{16}{R}$	Middle infra-red	14585 Å to 40515 Å
Pfund	5	6, 7, 8	$\frac{1}{\lambda_{56}} = R \left(\frac{1}{5^2} - \frac{1}{n^2} \right)$	$\frac{900}{11R}$	$\frac{25}{R}$	Far infra-red	22790 Å to 74583 Å

Energy levels A, B & C of a certain atom correspond to increasing values of energy, i.e. $E_A < E_B < E_C$. If $\lambda_1, \lambda_2, \lambda_3$ are the wavelengths of radiations corresponding to transitions C to B, B to A and C to A respectively then



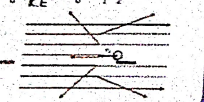
- a) $\lambda_3 = \lambda_1 + \lambda_2$ c) $\lambda_1 + \lambda_2 + \lambda_3 = 0$
b) $\lambda_3 = \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}$ d) $\lambda_3^2 = \lambda_1^2 + \lambda_2^2$

DISTANCE OF CLOSEST APPROACH OF α -PARTICLES

$$r_0 = \frac{1}{4\pi\epsilon_0} \frac{4Ze^2}{mv^2}$$

$$r_0 \propto \frac{1}{m} \quad r_0 \propto \frac{1}{v^2}$$

$$r_0 \propto \frac{1}{K.E} \quad r_0 \propto \frac{1}{Z_1 Z_2}$$



• RADIUS OF ORBIT $r_n = 0.53 \frac{n^2}{Z}$

• VELOCITY OF ELECTRON $v_n \propto \frac{Z}{n}$

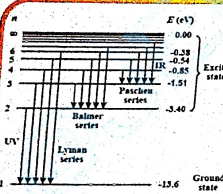
• TIME PERIOD $T \propto \frac{n^3}{Z^2}$

• FREQUENCY $\propto \frac{1}{T} \propto \frac{Z^2}{n^3}$

• CURRENT $\propto \frac{e}{T} \propto \frac{Z^2}{n^3}$

• MAGNETIC FIELD $B \propto \frac{v}{r^2} \propto B \propto \frac{Z^3}{n^3}$

• MAGNETIC DIPOLE MOMENT $M \propto n$



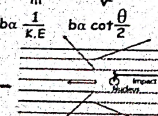
As the n value increases, the energy difference between adjacent level decreases
 $5 \rightarrow 1 > 4 \rightarrow 1 > 3 \rightarrow 1 > 2 \rightarrow 1 > 4 \rightarrow 2 > 3 \rightarrow 2$
 $E_{\text{min}} \propto \lambda_{\text{min}} \propto \lambda_{\text{max}}$

IMPACT PARAMETER

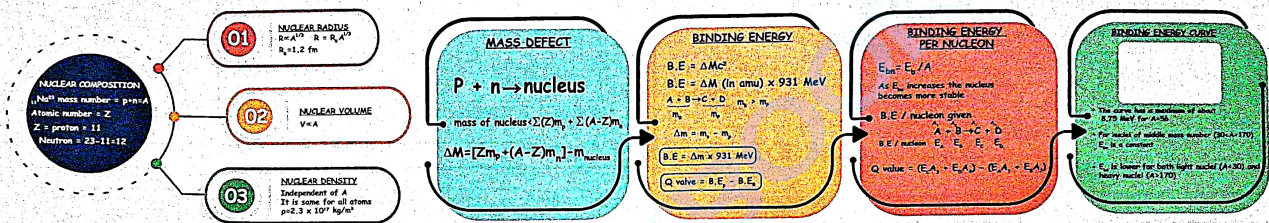
$$b = \frac{1}{4\pi\epsilon_0} \frac{Ze^2 \cot \frac{\theta}{2}}{\frac{1}{2} mv^2}$$

$$b \propto \frac{1}{m} \quad b \propto \frac{1}{v^2}$$

$$b \propto \frac{1}{K.E} \quad b \propto \cot \frac{\theta}{2}$$



NUCLEI



If radius of the ${}_{11}^{23}\text{Al}$ nucleus is taken to be R_u , then the radius of ${}_{52}^{131}\text{Te}$ nucleus is nearly
 (a) $\frac{23}{13} R_u$ (b) $\frac{5}{3} R_u$ (c) $\frac{3}{5} R_u$ (d) $\frac{131}{53} R_u$

NUCLEAR FORCE

01 Strongest force existing in nature



Nuclear force is short ranged
 $r < 0.8 \text{ fm}$ - attractive
 $r > 0.8 \text{ fm}$ - repulsive

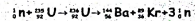
02

03 Nuclear force is charge independent
 $F_{p,p} = F_{p,n} = F_{n,n}$

Two deuterons undergo nuclear fusion to form a Helium nucleus. Energy released in this process is (given binding energy per nucleon for deuteron = 1.1 MeV and for helium = 7.0 MeV)
 (a) 30.2 MeV (b) 32.4 MeV (c) 23.6 MeV (d) 25.8 MeV

A radioactive nucleus (initial mass number A and atomic number Z) emits 3 α - particles and 2 positrons. The ratio of number of neutrons to that of protons in the final nucleus will be
 (a) $\frac{A-Z-8}{Z-4}$ (b) $\frac{A-Z-4}{Z-8}$ (c) $\frac{A-Z-12}{Z-4}$ (d) $\frac{A-Z-8}{Z-2}$

NUCLEAR FISSION



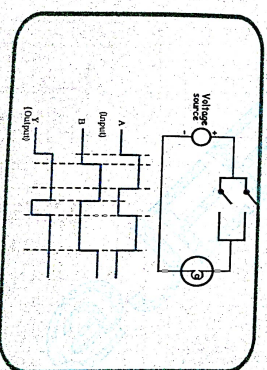
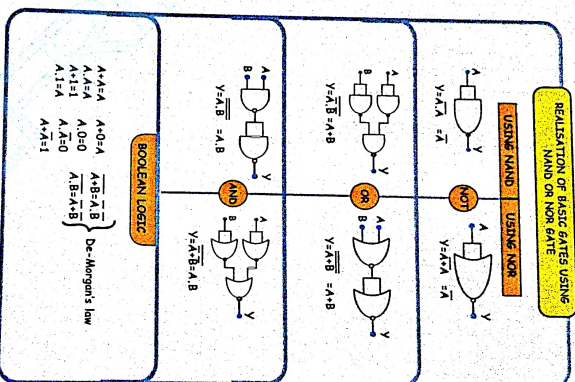
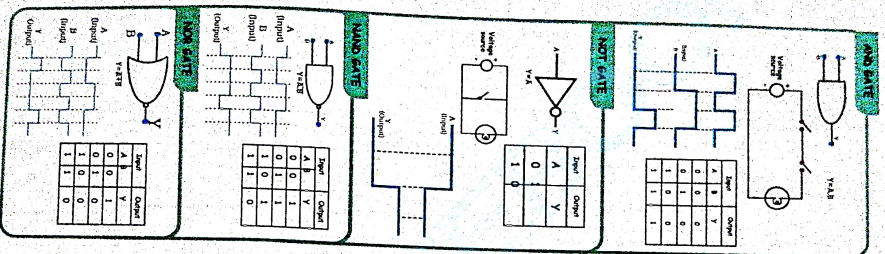
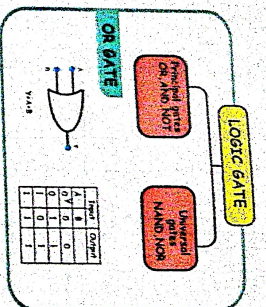
NUCLEAR REACTOR

- Multiplication Factor $k=1 \rightarrow$ critical
- Moderator : water, heavy water (D_2O), graphite and beryllium oxide.
- Control rods : Barium, cadmium
- Coolant : CO_2 , water, nitrogen

NUCLEAR FUSION

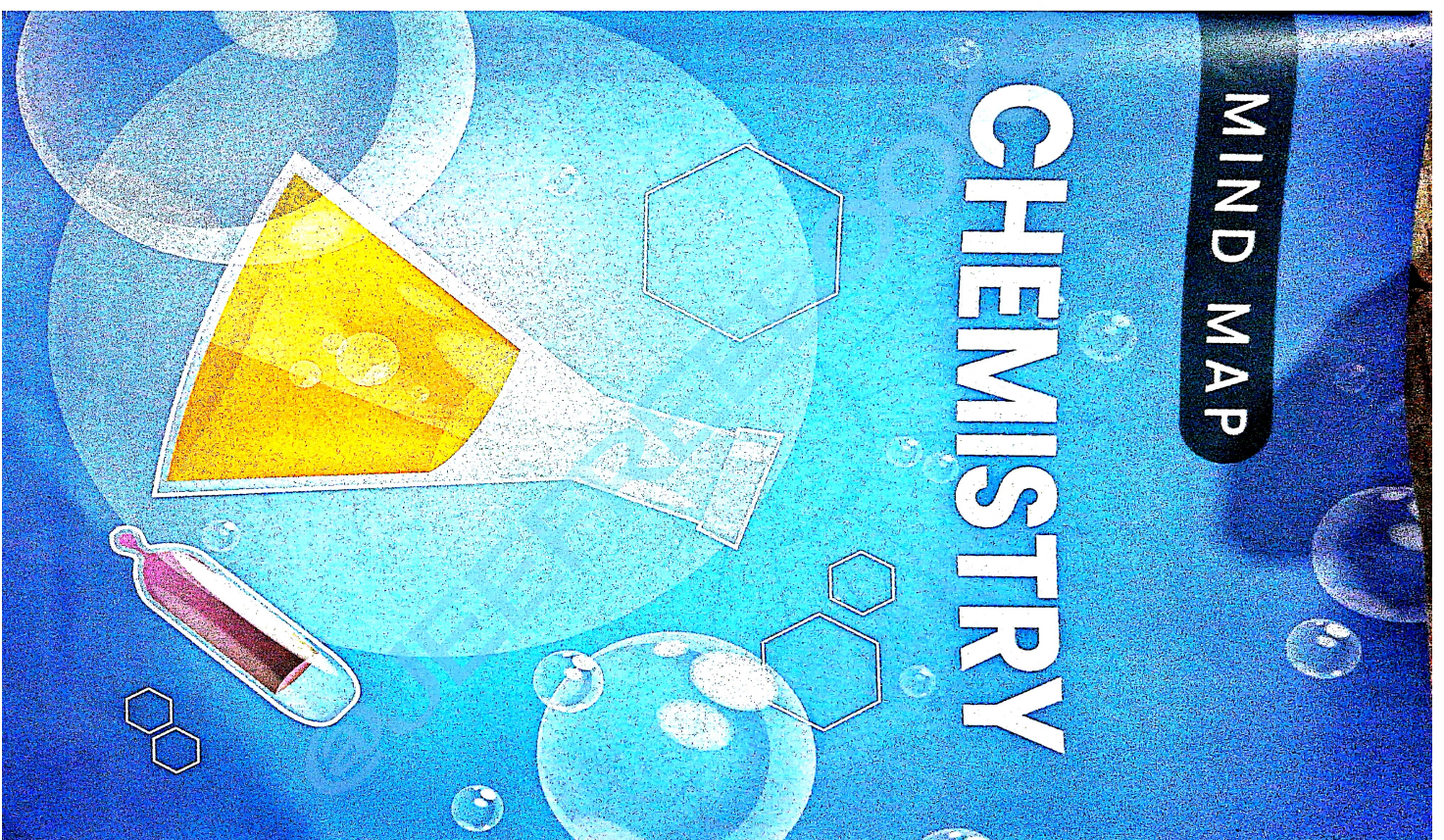
$4{}_1^1\text{H} + 2e^- \rightarrow {}_2^4\text{He} + 2\nu + 26.7 \text{ MeV}$
 Four hydrogen atoms combine to form an ${}^4\text{He}$ atom with the release of 26.7 MeV of energy.
 Achieved at very high temperature in order to overcome electrostatic repulsion.

ELECTRONICS

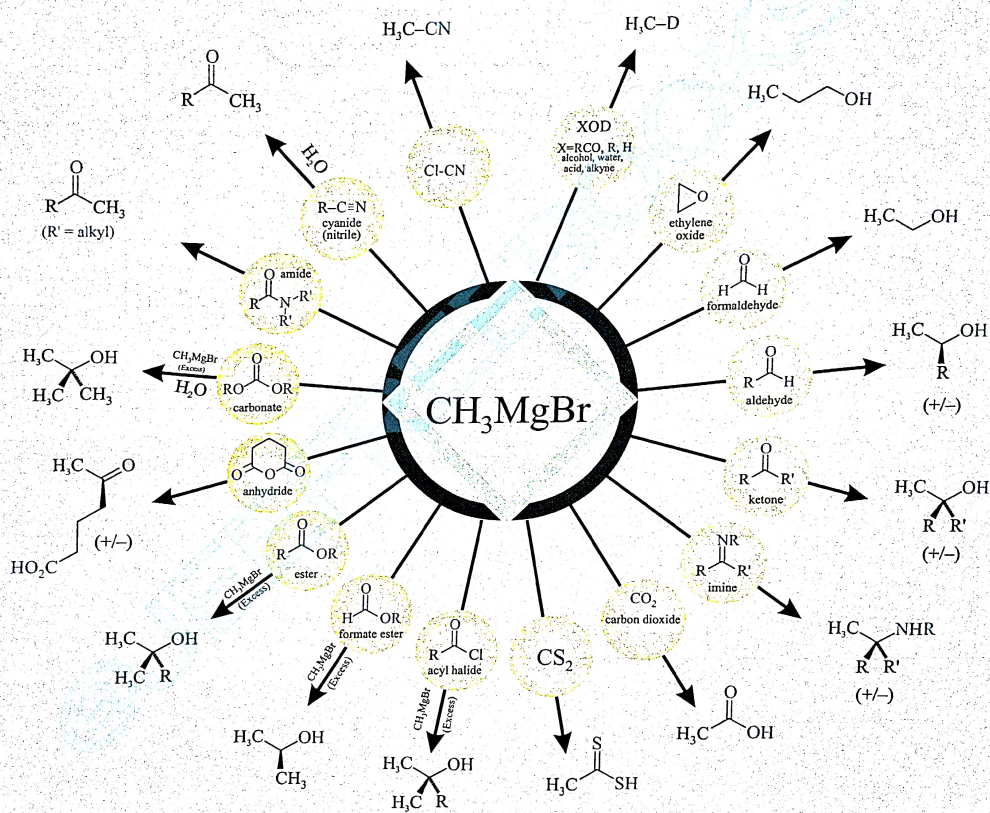


MIND MAP

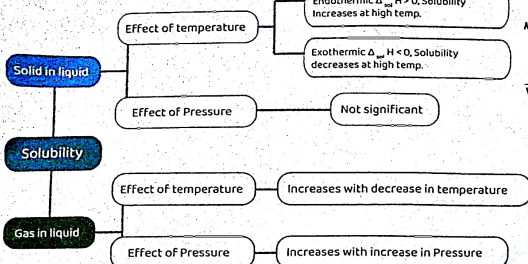
CHEMISTRY



GRIGNARD REAGENT C-C BOND



SOLUTIONS



Azeotropes:
The mixtures of liquids which boil at constant temperature like a pure liquid and possess same composition of components in liquid as well as vapour phase are called constant boiling mixtures or azeotropic mixtures.

Minimum boiling azeotropes:
They are formed by those liquid pairs which show positive deviations from ideal behaviour. eg: ethanol-water mixture.

Maximum boiling azeotropes:
They are formed by those liquid pairs which show negative deviations from ideal behaviour. eg: nitric acid-water mixture.

Colligative properties

Osmotic pressure $\rightarrow \pi = cRT$

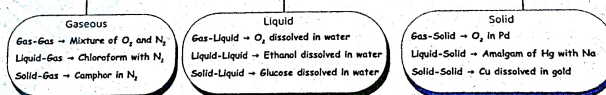
Depression in freezing point $\rightarrow \Delta T_f = K_f \times W_2 \times 1000$
 $M_2 \times W_1$

Elevation of boiling point $\rightarrow \Delta T_b = K_b \times 1000 \times W_2$
 $M_2 \times W_1$

Relative lowering of vapour pressure $\rightarrow \frac{W_2 \times M_1}{M_2 \times W_1} = \frac{P_1 - P_2}{P_1}$
(for dilute solutions)

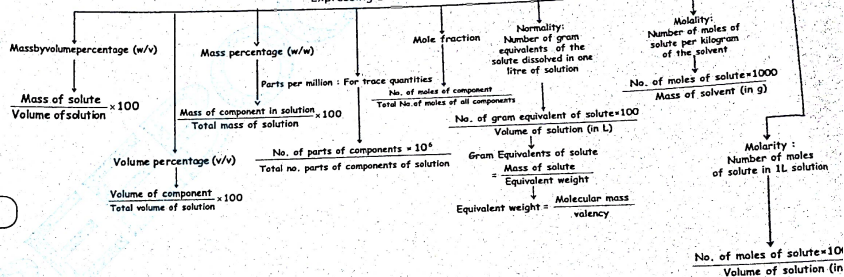
Henry's Law
Partial pressure of gas in vapour phase is proportional to the mole fraction of gas in the solution.
 $p = K_p \times x$

Types



Abnormal molecular mass
Molecular mass different from expected value
Van't Hoff factor (i) = $\frac{\text{Normal molar mass}}{\text{Abnormal molar mass}}$

Expressing Concentration of solutions



Raoult's Law
For any solution, the partial vapour pressure of each volatile component is directly proportional to its mole fraction in solution phase

Obey Raoult's law
Ideal solution $\rightarrow \Delta H_{\text{mix}} = \Delta V_{\text{mix}} = 0$
eg: n-hexane and n-heptane
Ethyl bromide + Ethyl chloride,
Chlorobenzene + Bromobenzene, etc.

Do not Obey Raoult's law
Non-Ideal solution $\rightarrow \Delta H_{\text{mix}} \text{ or } \Delta V_{\text{mix}} \neq 0$

Positive Deviation
 $\Delta H_{\text{mix}} > 0$
 $\Delta V_{\text{mix}} > 0$
eg: Acetone + Ethyl alcohol,
Water + Ethyl alcohol,
Carbon tetrachloride + Chloroform,
Chloroform + Ethanol,
Acetone + Carbon disulphide,
Acetone + Benzene, etc.

Negative Deviation
 $\Delta H_{\text{mix}} < 0$
 $\Delta V_{\text{mix}} < 0$
eg: Acetone + Aniline,
 $HCl + H_2O$, $HNO_3 + H_2O$,
 $H_2SO_4 + H_2O$,
Acetone + Chloroform,
 HNO_3 + Chloroform,
Benzene + Chloroform,
Pyridine + Glacial acetic acid, etc.

1 Electrochemical Cell

Left side	Right side
Oxidation	Reduction
Anode	Cathode
Negative	Positive

2 Representation of cell

Zn | Zn²⁺ || Cu²⁺ | Cu
R₁ | P₁ || R₂ | P₂

Product at anode Reactant at cathode

- Electrode potential (E_{cell})
E.P. = Reduction Potential (R.P)
= -Oxidation potential (O.P)
- If R.P = x, then O.P = -x
- Representation of Reduction half reaction:
 $M^{n+} + ne^- \rightarrow M$
- Standard Reduction Potential (SRP) (E_{SRP}°)
R.P at 1M and 298K.
SRP is calculated by using SHE
- Representation of SHE
 $H^+_{(aq)} | H_2 | Pt_{(s)}$ $E_{\text{SHE}}^{\circ} = 0$

3 Electrochemical series

Electrode	Standard Reduction Potential (E°)
Li ⁺ /Li	-3.04
K ⁺ /K	-2.93
Ca ²⁺ /Ca	-2.87
Na ⁺ /Na	-2.71
Mg ²⁺ /Mg	-2.37
Al ³⁺ /Al	-1.66
Zn ²⁺ /Zn	-0.76
Fe ²⁺ /Fe	-0.44
Sn ²⁺ /Sn	-0.14
Pb ²⁺ /Pb	-0.13
H ⁺ /H ₂	0.00
Cu ²⁺ /Cu	0.34
Ag ⁺ /Ag	0.80
Au ³⁺ /Au	1.68

SRP = O.A.
SRP = R.A.
Metals with high SRP = less reactive
Metals with low SRP = highly reactive

4 EMF of a cell

$$E_{\text{cell}}^{\circ} = E_{\text{Cathode}}^{\circ} - E_{\text{Anode}}^{\circ}$$

$$E_{\text{cell}} = E_{\text{Cathode}} - E_{\text{Anode}}$$

$$E_{\text{cell}} = E_{\text{Cathode}} + O.P.$$

$$E_{\text{cell}} = O.P. - E_{\text{Cathode}}$$

In cell, Cathode with high RP, Anode with low RP makes spontaneous reactions

5 Nernst equation

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.0591}{n} \log \frac{\text{Product}}{\text{Reactant}}$$

For Zn | Zn²⁺ || Cu²⁺ | Cu
 $E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.0591}{2} \log \frac{[Zn^{2+}]}{[Cu^{2+}]}$

For Ni | Ni²⁺ || Ag⁺ | Ag
 $E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.0591}{2} \log \frac{[Ni^{2+}]}{[Ag^+]^2}$

R₁ | P₁ || R₂ | P₂
If R₂ ↑, P₁ ↓ then E_{cell} ↑

6 Application of Nernst Equation

- Electrode Potential
 $E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.0591}{n} \log \frac{1}{[M^{n+}]}$
- Nernst equation in SHE
1) $E_{\text{cell}} = -0.0591 \log \frac{P_{H_2}}{[H^+]}$
If $P_{H_2} = 1 \text{ atm}$
(R.P.) = $E_{\text{cell}} = -0.0591 \text{ pH}$
(O.P.) = $E_{\text{cell}} = +0.0591 \text{ pH}$
- Concentration Cells
Zn | Zn²⁺ || Zn²⁺ | Zn
 $E_{\text{cell}} = \frac{0.0591}{n} \log \frac{[\text{cathode } C_1]}{[\text{anode } C_2]}$
 $\frac{C_1}{C_2} > 1 \Rightarrow \log \frac{C_1}{C_2} > 0 \Rightarrow E_{\text{cell}} > 0$

7 EMF; K_c & ΔG

$$E_{\text{cell}}^{\circ} = \frac{0.0591}{n} \log K_c$$

$$\Delta G = -nFE_{\text{cell}}$$

Spontaneous	Non-spontaneous
$\Delta G < 0$	$\Delta G > 0$
$E_{\text{cell}}^{\circ} > 0$	$E_{\text{cell}}^{\circ} < 0$
$\log K_c > 0$	$\log K_c < 0$
$K_c > 1$	$K_c < 1$

Galvanisation is applying coating of Zn

ELECTROCHEMISTRY

3 Faraday's law

Product formed

$$m = \frac{EM}{96500} \times I \times t$$

EM = $\frac{AM}{\text{valency}}$
1F = charge of 1 mole of e^- = 96500 C

Na⁺ + e^- → Na → 1F
Mg²⁺ + 2 e^- → Mg → 2F
Al³⁺ + 3 e^- → Al → 3F

1F displaces & gives 1 equivalent of product

1 mol = 23g of Na⁺
1 mol = 9g of Al³⁺
Ag⁺ → 108g → 1 mol

$O_2 \rightarrow 8g \rightarrow \frac{1}{4} \text{ mole} = 5.6L$
 $H_2 \rightarrow 1g \rightarrow \frac{1}{2} \text{ mole} = 11.2L$
 $Cl_2 \rightarrow 35.5g \rightarrow \frac{1}{2} \text{ mole} = 11.2L$
 $Cu^{2+} \rightarrow 31.75g \rightarrow \frac{1}{2} \text{ mole}$

4 Electrolytic conduction

Resistance (R) = $\frac{\rho l}{A}$ Unit of R = Ω
 $\rho = \Omega m$
Conductance (C) = $\frac{1}{R}$ C = $\Omega^{-1} = S = mho$
Conductivity (K) = $\frac{1}{\rho}$ K = $\Omega^{-1} m^{-1}$ or $S m^{-1}$
1 $S m^{-1}$ = 100 $S m^{-1}$

Molar Conductivity (λ_m)	Equivalent Conductivity (λ_{eq})
$\lambda_m = \frac{1000 K}{M}$	$\lambda_{eq} = \frac{1000 K}{N}$
K → $S cm^{-1}$	K → $S cm^{-1}$
M → $mol L^{-1}$	N → $eq L^{-1}$
$\lambda_m \rightarrow S cm^2 mol^{-1}$	$\lambda_{eq} \rightarrow S cm^2 eq^{-1}$

1 $S cm^2 mol^{-1}$ = $10^{-4} S m^2 mol^{-1}$
 $\lambda_m = \frac{1000}{M} \times \frac{1}{1000} = \frac{1}{M}$
N.M. : $\lambda_m \propto \frac{1}{M}$

For H_2SO_4 , Z = 2 (2H⁺)
NaCl, Z = 1 (1Na⁺)
Al₂(SO₄)₃, Z = 6 (2Al³⁺)

λ_m for SE increases with dilution (interionic attraction decreases)
At ∞ , $\lambda_m = \lambda_m^{\circ}$ (limiting molar conductivity)

of weak electrolytes increases with dilution (degree of dissociation increases)

strong electrolyte
weak electrolyte

λ_m° order: KCl > NaCl > LiCl
Conductance of Hydrated Rasil

1 Electrolytic cell

ANODE	CATHODE
Anion goes to anode	Cation goes to cathode
+ve electrode	-ve electrode
Oxidation	Reduction
A → A ⁺ + e^-	B + e^- → B ⁻
A → A ⁺ + ne^-	B ⁺ + ne^- → B

2 Product of electrolysis

Deposition order of cations (order of R.P)
Li⁺ < K⁺ < Ca²⁺ < Na⁺ < Mg²⁺ < Al³⁺ < Zn²⁺
Fe²⁺ < Ni²⁺ < H⁺ < Cu²⁺ < Hg²⁺ < Ag⁺ < Au³⁺

Deposition order of anion
SO₄²⁻ < NO₃⁻ < OH⁻ < Cl⁻ < Br⁻ < I⁻

Notes:

- For conc. H₂SO₄
Anode: H⁺ → 1 e^- + 1/2 H₂
Cathode: 2SO₄²⁻ + S₂O₈²⁻ + 2 e^- (peroxodisulphate ion)
- Very dil. NaOH (H₂O) + NaOH
Anode: H⁺ → 1 e^- + 1/2 H₂
Cathode: 2OH⁻ + 1/2 O₂ + H₂O + 2 e^-
- For CuSO₄ with Cu electrode
Anode: Cu → Cu²⁺ + 2 e^-
Cathode: Cu²⁺ + 2 e^- → Cu

Electroplating

5 Kohlrausch's law

$$\lambda_m^{\circ}(AB) = \lambda_m^{\circ}(A^+) + 2 \lambda_m^{\circ}(B^-)$$

$$\lambda_m^{\circ}(AB) = \lambda_m^{\circ}(A^+) + \lambda_m^{\circ}(B^-)$$

For Al₂(SO₄)₃
 $\lambda_m^{\circ}(Al_2(SO_4)_3) = 2 \lambda_m^{\circ}(Al^{3+}) + 3 \lambda_m^{\circ}(SO_4^{2-})$
 $\lambda_m^{\circ}(Al_2(SO_4)_3) = \lambda_m^{\circ}(Al^{3+}) + \lambda_m^{\circ}(SO_4^{2-})$

Application
 $\lambda_m^{\circ}(NH_4OH) = \lambda_m^{\circ}(NH_4^+) + \lambda_m^{\circ}(OH^-)$
 $\lambda_m^{\circ}(CH_3COOH) = \lambda_m^{\circ}(CH_3COO^-) + \lambda_m^{\circ}(H^+)$
 $\lambda_m^{\circ}(BaSO_4) = \lambda_m^{\circ}(Ba^{2+}) + \lambda_m^{\circ}(SO_4^{2-})$
 $\lambda_m^{\circ}(NaCl) = \lambda_m^{\circ}(Na^+) + \lambda_m^{\circ}(Cl^-)$